Effect of Low-level Image Processing on Satellite Images for Image Classification

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***Abstract*—Satellite image classification is an important topic in remote sensing applications, such as land use monitoring, environmental analysis and urban planning. On the contrary, classification models are extremely sensitive to input images as they often suffer from noise or low contrast and atmospheric disturbances. This paper studies low-level image preprocessing methods and their effect on the classification performance of satellite images. Preprocessing methods (Difference of Gaussian, Log Transform, Sobel filter Negative transformation, Laplacian filter, and Gamma correction) were applied on the EuroSAT dataset to two classes, AnnualCrop and PermanentCrop. For the evaluation, we used a ResNet50-based convolutional neural network with its classification head specifically trained. The experimental results show that the baseline (no preprocessing) and Difference of Gaussian methods obtained the highest classification accuracy with 95.10%. The Sobel filter completed the baseline model and achieved the highest precision of 97.20%,whenever, they further got best F1-score as well as recall with focus on balance between Precision & Recall which scored 95.20%. The result indicates that learning robust-feature representations using deep-learning models like ResNet50 works even without elaborate preprocessing, which means the gain of low-level features is limited and task-dependent in classifying satellite images.**

***Index Terms*—DoG, Laplacian, Sobel, Gamma Correction, ResNet50, TP, FP, TN, FN**

I. Introduction

Satellite imagery is essential to a number of remote sensing applications including land use and land cover monitoring, environmental assessment, urban planning, and disaster management. Considering the quickly evolving technologies for earth observations, daily, large amounts of satellite-based data enter into production systems, which allow researchers to study spatially and temporally dynamic environmental phenomena more conveniently [1]. Satellite images capture land cover and land use details of the Earth and image classification has been one of the common methods for extracting meaningful information from similar images by assigning one or more to a pixel or region in a satellite image [2]. But the efficacy of these classification systems is highly reliant upon the quality of your input images. Satellite images are generally raw in nature and have multiple problems such as noise, atmospheric distortion, illumination change, and low contrast. These conditions can result in image quality deterioration, and subsequent negative impact on machine learning and deep learning models performance for classification [3]. Hence, preprocessing is the initial step in satellite image analysis. Image preprocessing is the process of preparing the image for processing on an external stage by reducing noise, enhancing contrast, filtering the input and normalizing it to improve quality, and highlight important features in the image [4]. Low-level Image preprocessing is centered around simple pixel-level operations to improve visual appearance and structural domain properties of images. These are aimed at denoising steps, enhancing edge use and contrast, thereby helping the relevant features stand out more easily in subsequent analysis [5]. Such preprocessing methods are especially crucial in the case of satellite imagery as the images are acquired from very far, and they are quite susceptible to the effects of atmospheric conditions and sensor characteristics. Therefore, the performance of classification algorithms varies greatly with the application of preprocessing methods [6]. Even though machine learning and deep learning are being increasingly applied to satellite image classification, the effects of low-level preprocessing methods on classification performance have hardly been investigated in some studies. Gaining insight on the effect of specific preprocessing operations on classification accuracy will enable researchers to better design image analysis pipelines and increase the reproducibility and reliability of remote sensing applications [7]. This study investigates the impact of a number of low-level image preprocessing methods on classification performance of satellite images. Here the set of preprocessing methods, specifically Difference in Gaussian, Log Transform, Sobel, Negative, Laplacian and Gamma Corrections are applied to the Zenodo dataset [4] images before the classification. Experimental results present an understanding of the impact of the preprocessing strategies on the classification results and emphasize preprocessing as a principal stage in satellite image processing.

II. Dataset

The dataset used in this study is the EuroSAT Land Use and Land Cover Classification Dataset, a multispectral satellite image collection acquired from the Sentinel-2 platform under the Copernicus Programme. The original dataset comprises ten distinct land use and land cover classes; however, for the purpose of this study, a binary classification task was formulated by selecting two classes: AnnualCrop and PermanentCrop. Each selected class contains 2,500 images. Although the dataset provides multispectral information, only the visible spectrum (RGB bands) was utilized in this work. All images have a spatial resolution of 64×64 pixels with three channels. The dataset was partitioned into training, validation, and testing subsets using a 60:20:20 ratio. Specifically, 1,500 images per class were allocated for training, while 500 images per class were used for validation and 500 images per class for testing.

Fig. 1. Sample Data

III. Methodology

A. Preprocessing

Difference of Gaussians (DoG) is a band-pass filtering method that is constructed by subtracting two Gaussian-blurred images with distinct standard deviations. The initial Gaussian filter decreases the high-frequency noise and

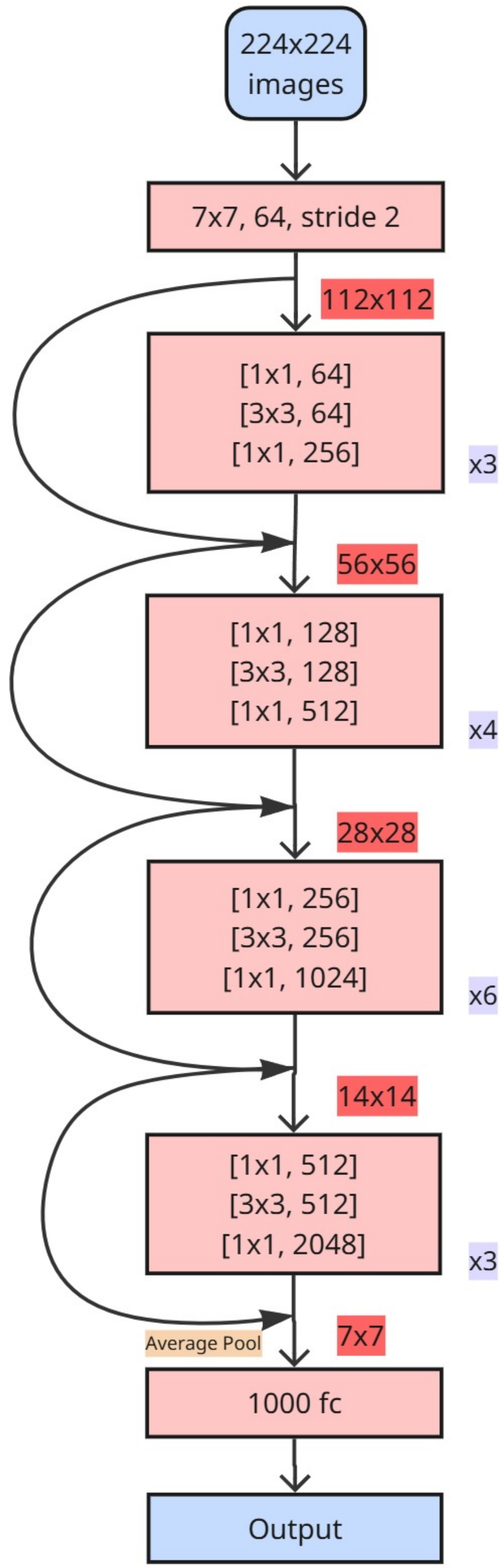

Fig. 2. ResNet50 Model Architecture

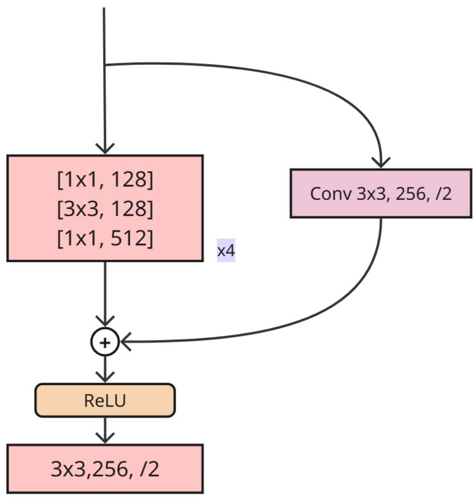

Fig. 3. Residual Block

smoothens the image, the second with higher standard deviations suppresses low-frequency components. When these two filtered images are subtracted, mid-frequency details are maintained and DoG is especially suitable for edge and feature enhancement.

Log transform is an intensity transfer method that highlights low pixel values (dark non-visible areas) and compresses high pixel values (bright non-visible areas) to make these regions more visible. It is a type of point processing, meaning that the transformation of each pixel is done independently of the other pixels. Logarithmic transformation used c = 1 in this study.

Sobel filter is a common high-pass filtering technique. It is applied in edge detection because it enhances high frequencies while leaving smooth regions alone. It works using the Sobel operator that calculates first-order derivatives of the image intensity, in horizontal and vertical directions. The last is then squared, summed, then square-rooted to form the magnitude of the gradient, the output coming from this is the edges within the image.

Negative image filter, is a point processing in which each pixel intensity is inverted. Thus, the bright areas turn to dark, and the dark areas turn to bright regions. While this transformation is quite simple, it is still popular because it can increase contrast visually and expose details that might otherwise be hidden, especially in the case of medical images or satellite images.

The 8-neighborhood Laplacian filter serves as a second-order derivative operator designed to identify edges by emphasizing areas with abrupt intensity shifts. In contrast to first-order approaches like Sobel, the Laplacian evaluates all eight adjacent pixels, diagonals included, to calculate the derivative. This comprehensive neighbor assessment increases sensitivity to subtle details and allows for the concurrent detection of edges across various orientations. Operating through a convolution mask, the filter calculates the sum of differences between a central pixel and its surroundings; this typically produces negligible output in consistent areas and high responses along edges. Due to its significant noise sensitivity, this filter is commonly implemented following a Gaussian smoothing process.

To modify image brightness, gamma correction utilizes a nonlinear intensity transformation based on a power-law relationship. This technique is frequently employed in display hardware and preprocessing workflows to adjust luminance and refine visual clarity. Gamma correction is often preferred over linear methods because it more closely corresponds to how the human eye perceives light and contrast. The primary distinctions between the 8-neighborhood Laplacian filter and gamma correction are found in their underlying mechanisms and objectives. While the Laplacian filter is a high-pass, spatial-domain technique used to highlight structural components through second-order derivatives, gamma correction is a point processing approach. The latter modifies individual pixel intensities to boost perceived brightness and contrast without evaluating neighboring data. Ultimately, the Laplacian is used for edge and detail enhancement, whereas gamma correction improves general image perception, making each suitable for specific objectives within the image processing pipeline.

B. Convolutional Neural Network

In deep learning, the two key components are feature extraction and classification. The feature extraction component is responsible for extracting meaningful representations from input images and passing them to the classification head, which then interprets those representations and produces a final output. For our proposed method, we selected ResNet50 as the backbone architecture and used a pre-trained version loaded with IMAGENET1K-V1 weights for fine-tuning. ResNet is a residual learning framework introduced to ease the training of very deep networks. It employs residual blocks with skip connections that allow gradients to flow directly across layers, mitigating the vanishing and exploding gradient problems that typically hinder deep network training. All backbone parameters were frozen during training, and the original fully connected classification head was replaced with a custom binary classification head. This head consists of a Linear layer (2048→512) followed by ReLU and Dropout (0.5), then a second Linear layer (512→256) with another ReLU and Dropout (0.3), and finally a Linear layer (256→1) with a Sigmoid activation to produce a binary output probability. Images were resized to 64×64 pixels and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). During training, random horizontal flipping and random rotation (±10°) were applied as data augmentation. Batches of 16 images were loaded using PyTorch's DataLoader with 2 worker processes. Binary Cross-Entropy Loss (BCELoss) was used as the loss function, suited for the binary classification task with a Sigmoid output. The optimizer was Adam, applied only to the unfrozen classification head parameters at a learning rate of 0.0001. A ReduceLROnPlateau

TABLE I
PERFORMANCE COMPARISON OF RESNET50 WITH DIFFERENT LOW-LEVEL FILTERS

Preprocessing	Accuracy	Precision	Recall	Specificity	F1 Score
None	95.10%	97.00%	93.45%	96.88%	95.20%
Difference in Gaussian	95.10%	96.60%	93.79%	96.49%	95.18%
Log Transform	94.00%	94.40%	93.65%	94.35%	94.02%
Sobel	94.40%	97.20%	92.05%	97.03%	94.56%
Negative	93.50%	96.20%	91.27%	95.98%	93.67%
Laplacian	90.60%	90.40%	90.70%	90.44%	90.58%
Gamma Correction	90.33%	92.53%	88.63%	92.19%	90.54%

scheduler was used, halving the learning rate when validation loss failed to improve for 2 consecutive epochs. Early stopping with a patience of 5 epochs was also applied, with the best model checkpoint saved based on minimum validation loss. Training was run for up to 30 epochs.

C. Evaluation Metrics

The proposed model was evaluated on classification metrics using a confusion matrix (TP, TN, FP and FN) including accuracy(1), precision(2), recall(3), specificity(4), and F1-score(5). Accuracy is a measure of overall correctness, precision indicates the confidence that AnnualCrop predictions are correct, and recall measures the model's ability to capture all AnnualCrop samples. The specificity measures the ability to identify PermanentCrop samples, and F1-score is a combination of precision and recall, which is well balanced for the high imbalanced datasets usually present in image analysis.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{4}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

IV. RESULT AND DISCUSSION

The results of the ResNet50 model with the different preprocessing conditions are in Table 1. In summary, we have robust and consistent results from the model with the best accuracy, 95.10%, when no filter and Difference of Gaussian filtering are used. As seen in the baseline (no filter) setting, the model obtains a precision of 97.00%, a recall of 93.45%, a specificity of 96.88%, and an F1 score of 95.20%, which reflects a balance between false positives and false negatives. Applying some preprocessing slightly alters the performance; the DoG filter works reasonably well, and the Log Transform reduces overall classification accuracy and balance, pointing to a loss of useful feature information. The results of the Sobel filter precision is 97.20% and specificity is 97.03%, while recall is 92.05%, which shows that the model is more conservative to detect positive cases. Of all the variants, the Negative transformation demonstrates the least performance overall with lowered recall and F1 score. The above results indicate that even though preprocessing has an effect on feature extraction, the model seems to already learn strong representations without any heavy transformations. This follows the somewhat generic advice in deep learning in that architectures such as ResNet utilize hierarchical feature learning, in that early layers capture edges and textures (like Sobel), and deeper layers learn more abstract patterns, meaning that the need for complete manual preprocessing is not as important when using strong pre-trained or deep architectures.

V. CONCLUSION

This work aims to compare the effect of low-level preprocessing techniques to classify satellite images. For classifying the dataset we used ResNet50 architecture. We trained multiple models with the same architecture on datasets filtered with various filtering techniques. Conducting evaluations, we found that the Difference of Gaussian (DoG) preprocessing achieves the highest classification accuracy (95.1%) and also the highest recall (93.79%) amongst all evaluated methods. Of all datasets, the Sobel-filtered dataset achieved the greatest precision (97.2%; Sensitivity 94.1%; Specificity 97.03%). On the other hand, the baseline model without preprocessing produced the highest F1-score, implying the model performed well on both precision and recall. All the models ran for 30 epochs and early stopping was enabled where patience was set 5 epochs.

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